5 An alternating supply of frequency 50 Hz and output 6.0 V r.m.s. is rectified to provide direct current for a resistor R, as shown in Fig. 5.1.

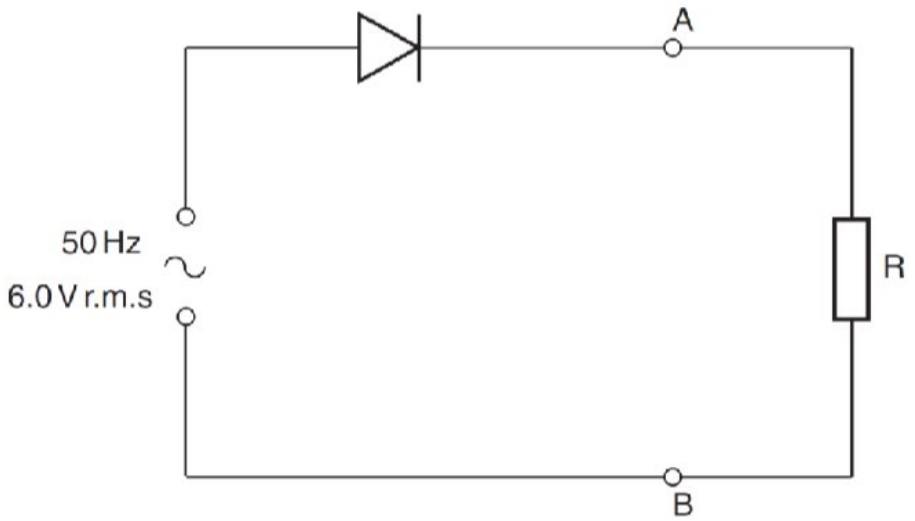


Fig. 5.1

The diode is ideal.

The output potential difference V of the alternating power supply is represented by

$$V = V_o \sin(\omega t)$$

where V_o is the maximum potential difference in volts, ω is the angular velocity in rad s^{-1} and t is the time in seconds.

(a) By reference to heating effect, state what is meant by the *root-mean-square (r.m.s.)* of an alternating current.

.....

.....

.....[1]

(b) Determine the value of V_o.

V_o = ... V [1]

(c) Determine the value of ω.

value of ω = ...rad s^{-1} [1]

(d) Determine the maximum potential difference across the diode during one cycle.

potential difference = ... V [1]

(e) State the potential difference across A and B when the diode has maximum potential difference across it. Give a reason for your answer.

.....

..... [1]

(f) R has a resistance of 5.0 Ω .

In Fig. 5.2, sketch the variation with time of the power dissipated in R for **one** period of the alternating supply. Show the maximum power dissipated in R clearly in your graph.

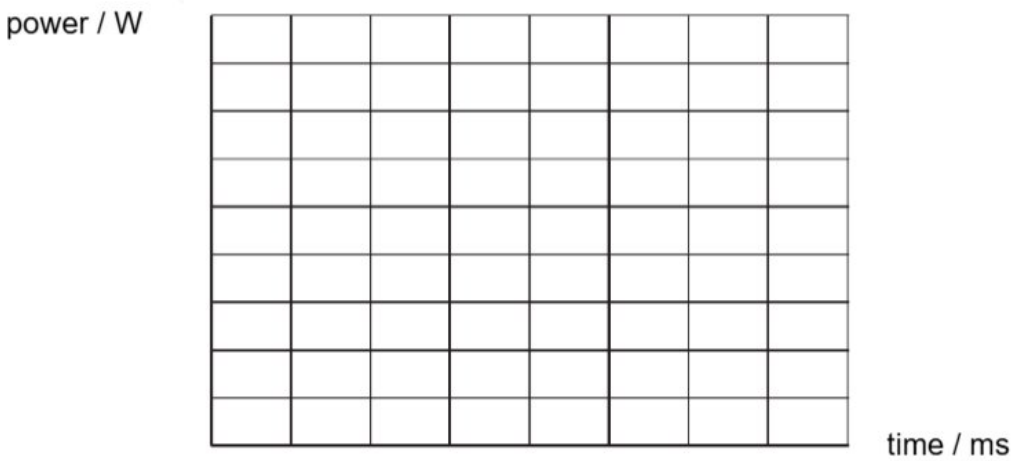


Fig. 5.2

[2]

[Total: 7]